

Auto Power-On & AC Recovery Mechanism Analysis

Hardware Profile: Intel NUC6i7KYB | Telemetry & BIOS Diagnostics | Generated: 2026-08-31 12:05:45 BST

Executive Summary & Diagnostic Verdict:

Your system is identified as an **Intel NUC6i7KYB (Skull Canyon)**. The Windows Event logs confirm that on **August 30 at 08:51 UTC** and **August 31 at 06:44 UTC**, the system experienced sudden unexpected power cycles (Kernel-Power Event ID 41 & 6008) and **did automatically boot back up** once power was re-applied. During the preceding inactive period (Aug 24–29), the system did not power on because either AC line power was never interrupted, or the BIOS is configured to **'Last State'** (which keeps the unit OFF if it was previously shut down).

1. How 'Restore on AC Power Loss' Operates

In modern PC and Intel NUC architectures, the **'After Power Failure'** (AC Recovery) setting is handled by the hardware Power Management Controller (PMC) and Super I/O chipset. It only triggers when electrical mains power transitions from **G3 Mechanical Off (0 Volts AC/DC)** to **AC Connected**. There are three key configuration modes:

BIOS Setting Mode	Behavior on Power Restoration	Operational Assessment
Power On / Always On	System powers on immediately whenever AC power is restored, regardless of prior state.	Recommended for autonomous 24/7 headless servers, remote access, and unattended operation.
Last State / Previous State (Often Factory Default)	System powers ON if it was running when power was lost. If it was shut down cleanly (S5), it remains OFF .	If you shut down the machine and later toggle power, it will NOT turn on because its recorded last state was OFF.
Stay Off / Power Off	System always stays OFF after power restoration until the physical button is pressed.	Default desktop safety mode to prevent unexpected startup after electrical grid restoration.

2. Root Cause Analysis: Why Was the System Offline Aug 24–29?

Possibility A: 'Last State' Mode was Active After a Shutdown

On August 23 at 00:02 UTC, the system initiated a shutdown/restart sequence. If the system was left in a powered-down state (S5) and the BIOS setting was 'Last State', cutting and restoring wall power during the week would leave the system OFF.

Possibility B: Continuous Wall Power (No AC Loss Occurred)

The 'After Power Failure' circuitry only activates upon an actual loss and restoration of AC voltage. If mains power to the NUC power brick was uninterrupted throughout the workweek, no trigger condition occurred to initiate an automatic boot.

Possibility C: 19V DC Power Brick Capacitor Hold-Up

Intel NUC devices use an external 19V DC power adapter. During brief brownouts or momentary power flickers (< 2 seconds), the capacitors in the power brick maintain DC voltage above the G3 trip threshold, preventing the motherboard from detecting an AC failure cycle.

3. Step-by-Step BIOS Configuration for Intel NUC6i7KYB

To guarantee that your Intel NUC powers on 100% of the time whenever power is connected or restored, verify these BIOS settings:

- Access BIOS:** Power on the Intel NUC and press **F2** repeatedly during startup to enter Intel Visual BIOS.
- Navigate to Power Settings:** Click the **Advanced** tab > Select **Power** from the top menu > Select **Secondary Power Settings**.
- Configure After Power Failure:** Change 'After Power Failure' from Last State to **Power On**.
- Disable Deep S4/S5:** In the same menu, ensure 'Deep S4/S5 State' is set to **Disabled** (allows remote wake / smart plug control).
- Save & Exit:** Press **F10** and select **Save Changes and Exit**.